Machine Learning & Deep Learning

Experiment 6

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AIM: Apply K-Means and Hierarchical Clustering on sample datasets

1. Dataset Source

- **Dataset Name:** Mall Customer Segmentation Data
- **Source:** **Kaggle - Mall Customer Segmentation Data**

2. Dataset Description

This dataset contains data on customers of a mall. The aim is to segment customers into distinct groups based on their spending behavior and income to aid marketing strategies.

- **Size:** 200 records, 5 columns.
- **Target Variable:** None (Unsupervised Learning).
- **Features:**
  - **CustomerID** : Unique ID.
  - **Gender** : Male/Female.
  - **Age** : Age of the customer.
  - **Annual Income** (k$) : Annual Income of the customer.
  - **Spending Score (1-100)** : Score assigned by the mall based on customer behavior and spending nature.

3. Mathematical Formulation of the Algorithm

A. K-Means Clustering:

K-Means partitions n observations into k clusters in which each observation belongs to the cluster with the nearest mean (centroid).

- **Objective:** Minimize the **Within-Cluster Sum of Squares (WCSS)** , also known as Inertia.

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} ||x_i^{(j)} - c_j||^2$$

Where:

- $x_i^{(j)}$ is a data point in cluster j.
- c_j is the centroid of cluster j.
- $||...||^2$ is the squared Euclidean distance.

B. Hierarchical Clustering (Agglomerative):

This is a "bottom-up" approach.

1. Start with n clusters (each data point is its own cluster).
2. Find the two closest clusters and merge them.
3. Repeat until only one cluster remains.

- **Linkage Criteria:** Determines distance between sets of observations. **Ward's Method** is commonly used, which minimizes the variance within clusters being merged.

4. Algorithm Limitations

- **K-Means:**
  - 0 **Specify K:** You must define the number of clusters (k) beforehand.
  - ○ **Spherical Assumption:** Assumes clusters are spherical and of similar size. It fails with complex shapes (e.g., concentric circles).
  - ○ **Sensitive to Outliers:** Centroids can be dragged by outliers.

- **Hierarchical:**
  - 0 **Computationally Expensive:** Time complexity is high ($O(n^3)$), making it unsuitable for very large datasets.
  - ○ **Irreversible:** Once a step (merge) is done, it cannot be undone.

5. Methodology / Workflow

1. **Data Loading:** Load Mall_Customers.csv.

```
# 1. Load Dataset try:
    df = pd.read_csv('Mall_Customers.csv') print("Dataset loaded
    successfully.")
except FileNotFoundError: print("Error: 'Mall_Customers.csv' not found. Please upload the file.") df
    = pd.DataFrame()
```

2. **Feature Selection:** Isolate Annual Income and Spending Score for clear 2D visualization.

```
# 2. Feature Selection
# We select Annual Income (Index 3) and Spending Score (Index 4) X = df.iloc[:, [3,
    4]].values
```

3. **K-Means Workflow:**
   - Use the **Elbow Method** (plot WCSS vs K) to find optimal K.
   - Train K-Means model.
   - Visualize clusters.

```
    # 1. Elbow Method to find optimal K wcss = []
for i in range(1, 11):
        kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42, n_init=10)
kmeans.fit(X) wcss.append(kmeans.inertia_)

plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), wcss, marker='o', color='red') plt.title('The Elbow
Method') plt.xlabel('Number of clusters') plt.ylabel('WCSS') plt.show()

# 2. Train K-Means with Optimal K (Usually 5 for this dataset) k_optimal = 5
kmeans = KMeans(n_clusters=k_optimal, init='k-means++', random_state=42, n_init=10)
y_kmeans = kmeans.fit_predict(X)

# 3. Visualizing the Clusters plt.figure(figsize=(10, 6)) colors = ['red',
'blue', 'green', 'cyan', 'magenta'] for i in range(k_optimal):
        plt.scatter(X[y_kmeans == i, 0], X[y_kmeans == i, 1], s=100, c=colors[i],
                    label=f'Cluster {i+1}')

# Plot Centroids plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], s=300,
c='yellow', label='Centroids', edgecolors='black')
plt.title('K-Means: Clusters of Customers')
plt.xlabel('Annual Income (k$)') plt.ylabel('Spending Score
(1-100)') plt.legend() plt.show()
```

4. **Hierarchical Workflow:**
   - Plot a **Dendrogram** to visually determine the optimal number of clusters.
   - Train Agglomerative Clustering model.
   - Visualize clusters.

```
# ==========================================
# PART B: HIERARCHICAL CLUSTERING
# ========================================== print("\n--- Part B:
Hierarchical Clustering ---")

# 1. Dendrogram to find optimal clusters plt.figure(figsize=(10, 6))
dendrogram = sch.dendrogram(sch.linkage(X, method='ward'))
plt.title('Dendrogram') plt.xlabel('Customers') plt.ylabel('Euclidean Distances')
plt.show()

# 2. Train Agglomerative Clustering (Using K=5 based on Dendrogram) hc =
AgglomerativeClustering(n_clusters=5, metric='euclidean', linkage='ward')
y_hc = hc.fit_predict(X)

# 3. Visualizing the Clusters
plt.figure(figsize=(10, 6)) for i in range(5):
    plt.scatter(X[y_hc == i, 0], X[y_hc == i, 1], s=100, c=colors[i],
                label=f'Cluster {i+1}')

plt.title('Hierarchical: Clusters of Customers') plt.xlabel('Annual
Income (k$)') plt.ylabel('Spending Score (1-100)') plt.legend()
plt.show()
```

5. **Evaluation:** Calculate **Silhouette Score** (measures how similar an object is to its own cluster compared to other clusters).

```
# 4. Evaluation score_kmeans = silhouette_score(X, y_kmeans) print(f"
K-Means Silhouette Score : {score_kmeans:.4f}")

# 4. Evaluation score_hc = silhouette_score(X,
y_hc)
print(f" Hierarchical Silhouette Score : {score_hc:.4f}")
```

6. **Performance Analysis**

- **Metric:** Silhouette Coefficient (Range: -1 to +1).

- **Interpretation:** A score closer to +1 indicates that data points are well-clustered (dense and well-separated). A score near 0 indicates overlapping clusters.

Code:

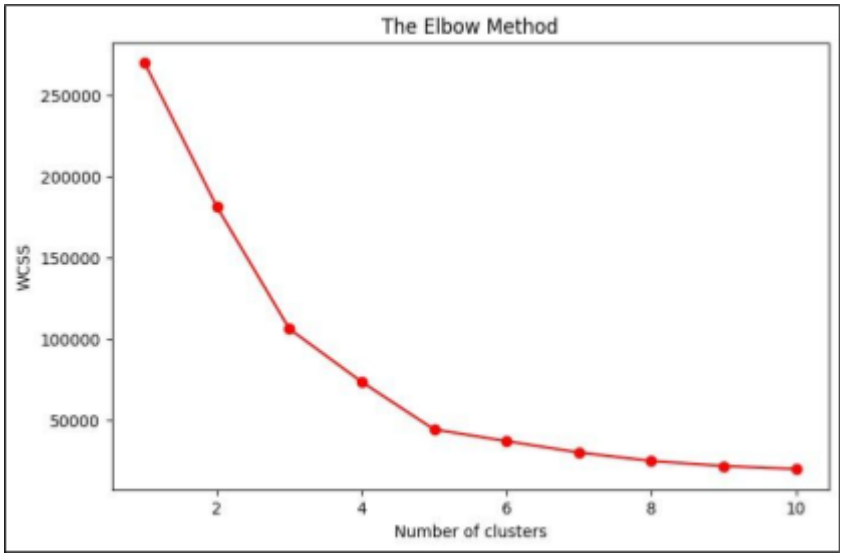
```
import pandas as pd
import numpy as np
 import matplotlib.pyplot as plt import seaborn as
sns from sklearn.cluster import KMeans,
AgglomerativeClustering from sklearn.metrics import
silhouette_score import scipy.cluster.hierarchy as sch
print ( "--- EXPERIMENT 6: CLUSTERING ALGORITHMS ---" )

 try :
     df = pd.read_csv( 'Mall_Customers.csv' )
    print ( "Dataset loaded successfully." )
 except FileNotFoundError: print ( "Error: 'Mall_Customers.csv' not
    found. Please upload the file." ) df = pd.DataFrame()

 if not df.empty:
     X = df.iloc[:, [3 , 4 ]].values

    print ( "\n--- Part A: K-Means Clustering ---"
    )
     # 1. Elbow Method to find optimal K wcss = [] for i in range ( 1 , 11 ):
    kmeans = KMeans(n_clusters=i, init= 'k-means++' , random_state= 42 , n_init=
    10) kmeans.fit(X)
         wcss.append(kmeans.inertia_)

     plt.figure(figsize=( 8 , 5 )) plt.plot( range ( 1
    , 11 ), wcss, marker= 'o' , color= 'red' )
    plt.title( 'The Elbow Method' ) plt.xlabel( 'Number
    of clusters' ) plt.ylabel( 'WCSS' ) plt.show()
    k_optimal = 5

     kmeans = KMeans(n_clusters=k_optimal, init= 'k-means++' , random_state=
    42 ,
 n_init= 10 ) y_kmeans =
    kmeans.fit_predict(X)
```

```
 plt.figure(figsize=( 10 ,  6 ))  colors = [ 'red',
'blue' ,  'green' ,  'cyan' ,  'magenta' ]  for  i
in  range (k_optimal):
     plt.scatter(X[y_kmeans == i,  0 ], X[y_kmeans  == i,
                1 ],  s= 100 , c=colors[i], label= f
                'Cluster  {i+ 1 } ' )

 plt.scatter(kmeans.cluster_centers_[:, 0  ], kmeans.cluster_centers_[:, 1
            ],  s= 300 , c= 'yellow' , label= 'Centroids' ,  edgecolors=
            'black' )
 plt.title( 'K-Means: Clusters of
Customers' )  plt.xlabel( 'Annual Income
(k$)')   plt.ylabel( 'Spending Score
(1-100)')   plt.legend()  plt.show()

score_kmeans = silhouette_score(X, y_kmeans)  print (
f "K-Means Silhouette Score:  {score_kmeans :.4f } " )


  print ( "\n--- Part B: Hierarchical Clustering
---")

plt.figure(figsize=( 10,  6  ))  dendrogram =
sch.dendrogram(sch.linkage(X, method= 'ward' ))
plt.title( 'Dendrogram')   plt.xlabel( 'Customers')
plt.ylabel( 'Euclidean Distances' )  plt.show()

hc = AgglomerativeClustering(n_clusters=5  , metric= 'euclidean',   linkage=
'ward')   y_hc = hc.fit_predict(X)

 plt.figure(figsize=( 10 ,  6
))  for  i  in  range ( 5 ):
     plt.scatter(X[y_hc == i,  0 ], X[y_hc == i, 1  ],
            s= 100 , c=colors[i], label= f 'Cluster  {i+ 1 } ')

 plt.title( 'Hierarchical: Clusters of
Customers' )  plt.xlabel( 'Annual Income (k$)' )
plt.ylabel( 'Spending Score (1-100)' )
plt.legend()  plt.show()
```

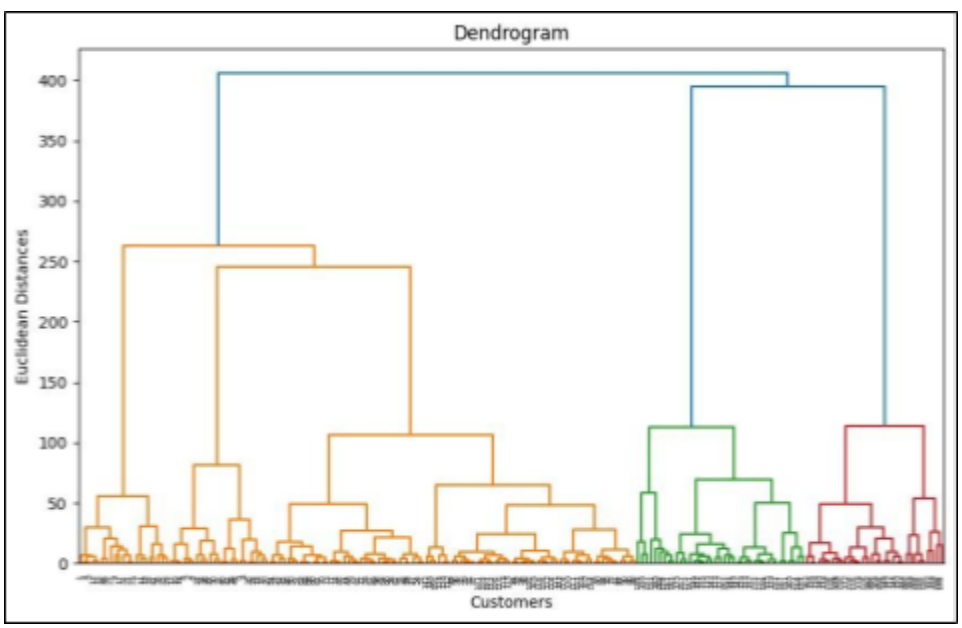
```
score_hc = silhouette_score(X,
y_hc)
 print ( f "Hierarchical Silhouette Score:  {score_hc :.4f}  ")
```

7. Hyperparameter Tuning

- **K-Means:** Tuning K (Number of clusters). We look for the "elbow" point where the drop in WCSS slows down.



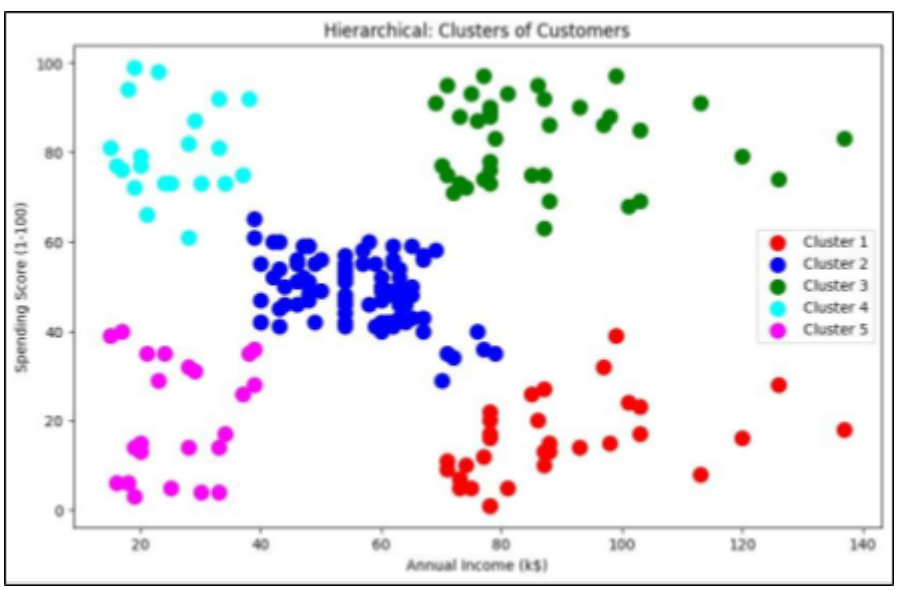
- **Hierarchical:** Tuning the "Cut-off" point on the Dendrogram to decide how many clusters to form.



Output:

```
--- EXPERIMENT 6: CLUSTERING ALGORITHMS ---
Dataset loaded successfully.

Part A: K-Means Clustering
```

```
K-Means Silhouette Score: 0.5539
Part B: Hierarchical Clustering
```




```
Hierarchical Silhouette Score: 0.5530
```

8. Conclusion

In this final experiment, we applied unsupervised learning techniques to segment customers.

- **The Elbow Method** and **Dendrogram** both clearly indicated that **5** is the optimal number of clusters for this dataset.

- **Cluster Interpretation:** The algorithms successfully identified 5 distinct customer profiles:
  1. **Savers:** High Income, Low Spending.
  2. **Spenders:** High Income, High Spending (Target Audience).
  3. **Careful:** Low Income, Low Spending.
  4. **Careless:** Low Income, High Spending.
  5. **Average:** Moderate Income, Moderate Spending.

- **Comparison:** Both K-Means and Hierarchical Clustering produced nearly identical results on this dataset, as confirmed by similar Silhouette Scores. K-Means is generally preferred for larger datasets due to speed, while Hierarchical is better for understanding the nested relationships in smaller datasets.